

Excel Data Tools — Detailed Lecture Notes

JKSSB Computer Awareness

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27.1 What Excel data tools do

Once data has been entered into a worksheet, it usually has to be organised, narrowed, summarised, and presented. Excel groups the features that do this under what we can call its data tools: **sort**, **filter**, **tables**, **charts**, **PivotTables** and **PivotCharts**, **printing**, **protection**, and **what-if analysis**. These are the features a JKSSB candidate is expected to recognise by name and by one-line purpose.

Definition (Data tools). *The worksheet features that arrange, narrow, summarise, present, print, and safeguard data without changing the underlying values.*

The single idea that holds the whole lesson together is this: most of these tools change the *view* of the data. Sorting, filtering, and a PivotTable reorganise what you see on the screen; the original values in the cells stay exactly where they were. The exceptions are deliberate edits — a sort does physically move rows, and a user can type new data — but the summary tools themselves are read-only with respect to the source.

Tool	One-line purpose
Sort	Arrange rows in a chosen order.
Filter	Show only the rows that meet a condition.
Excel table	A named, auto-expanding, formatted range of data.
Chart	A picture of the data, linked to its source range.
PivotTable	A fast summary of a large table of data.
PivotChart	A chart built on a PivotTable.
Protection	Restrict editing of cells or workbook structure.
What-if analysis	Test how the result changes with different inputs.

27.2 Sorting

Sorting rearranges rows into an order chosen by the user. **Ascending** order goes from A to Z, or from the smallest number to the largest. **Descending** order goes from Z to A, or from the largest number to the smallest. A sort is performed on a *column*, but the whole row moves with that column so that each record stays intact.

The **Sort** dialog (**Data > Sort**) can order data by **more than one level**. For example, a payroll sheet can be sorted by Department first, and then by Salary within each department. The dialog's **Add Level** button adds these secondary and tertiary keys.

A **custom list** sorts by a defined order rather than alphabetically: days of the week and months are the standard examples. Excel can also sort by cell colour, font colour, or cell icon. When the

range has titles, the option **My data has headers** must be ticked so that the header row is not sorted along with the data.

Example 27.2.1. *A teacher’s marksheet has the columns Roll No., Name, and Marks. Clicking anywhere in the Marks column and choosing Sort Largest to Smallest reorders the rows so the highest marks appear first. If only the Marks column is selected and the rest of the row does not move, the records are scrambled — the names no longer match the marks.*

Trap watch. Sorting a **single column** on its own scrambles the records. Always sort the whole table, or convert the data to an Excel table first.

27.3 Filtering

Filtering displays only the rows that satisfy a condition and hides the rest. The hidden rows are **not deleted**; they are temporarily out of sight. AutoFilter is switched on and off with **Data > Filter**, whose keyboard shortcut is **Ctrl+Shift+L**. Once it is on, drop-down arrows appear in the header row of each column.

Definition (Filter). *A tool that shows only the rows matching a condition while hiding — not removing — the other rows.*

Clicking a drop-down arrow allows two broad kinds of narrowing:

- a **value filter**, where one or more ticks are kept from a list of the column’s values;
- a **condition filter**, where a rule is applied. Text Filters, Number Filters, and Date Filters offer common rules such as *Equals*, *Greater Than*, *Begins With*, and *Between*.

Top 10 and **Above Average** are ready-made condition filters. **Advanced Filter** is used for more complex criteria and can copy the matching rows to another location on the sheet.

Clearing the filter with **Clear Filter** restores every row. The key contrast to remember is that filtering *hides* rows; it does not remove them and does not change the data.

Example 27.3.1. *A sales sheet has a Region column. Applying a value filter and ticking only “North” hides all other regions. The rows for South, East, and West are still present in the worksheet; clearing the filter brings them back unchanged.*

27.4 Excel tables

An **Excel table** is a range of data that has been converted with **Insert > Table** (shortcut **Ctrl+T**) and given a table style. A table has a **name** (by default *Table1*), and it behaves as a single object rather than as loose cells.

Definition (Excel table). *A named, formatted range of data that automatically expands to include new rows and columns and whose formulas use structured references.*

The practical advantages of a table over a plain range are:

- it **auto-expands** when a new row is typed immediately below it;
- it shows **banded rows** that improve readability;
- it adds **filter buttons** to the header row automatically;

- it can display an optional **Total Row**; and
- its formulas use **structured references** such as **Table1[Sales]** instead of cell addresses such as **B2**.

Because a structured reference points at the whole column, it automatically covers new rows that are added to the table. A formula written with cell addresses would have to be extended by hand.

Point	Plain range	Excel table
Name	None	Assigned name (Table1)
Grows	Manually	Auto-expands
Formulas	Cell addresses	Structured references
Filters	Manual AutoFilter	Built-in header buttons

A table is *not* a PivotTable. A table is the **source data**; a PivotTable is one of the summaries that can be built on top of it.

27.5 Charts

A **chart** is a graphical picture of worksheet data. Because the chart is **linked** to its source range, editing the numbers in the cells updates the chart automatically. The common chart types and their uses are:

Chart type	Best used for
Column / bar	Comparing values across categories.
Line	Showing a trend, usually over time.
Pie / doughnut	Showing the parts of a single whole.
Scatter (XY)	Showing the relationship between two numeric variables.

The parts of a chart carry their own names, and exam items often test them. The **chart area** is the whole chart, including the frame. Inside it, the **plot area** is the region where the data is actually drawn. Each set of values is a **data series**, and the **legend** names each series. The **category axis** (usually the X-axis) carries the labels, while the **value axis** (the Y-axis) carries the numbers. **Data labels** display the exact value at each data point, and **gridlines** help read the values.

A chart normally floats on the same worksheet as its data, but it can be moved to a separate **chart sheet** that contains only the chart.

Example 27.5.1. *A school records the number of students in each class. A column chart compares the classes at a glance. The same data as a line chart would suggest a trend across the classes, which may not be meaningful; for parts of one total, such as the share of boys and girls in the school, a pie chart is the right choice.*

27.6 PivotTables and PivotCharts

A **PivotTable** summarises a large table of data in seconds. Instead of writing formulas, the user **drags** fields into four areas:

Area	What goes there
Rows	Field values listed down the left side of the summary.
Columns	Field values spread across the top of the summary.
Values	The numbers that are summarised (Sum, Count, Average, and more).
Filters	Fields used to filter the whole PivotTable.

The **Values** area chooses the aggregation: by default a numeric field is summed and a text field is counted, but Count, Average, Max, Min, and other functions can be chosen. Rearranging the field layout re-computes the summary instantly — this quick re-orientation is what gives the PivotTable its name.

Definition (PivotTable). *An interactive summary that aggregates a large range of data by dragging fields into Rows, Columns, Values, and Filters areas.*

A **PivotChart** is a chart built on a PivotTable and linked to it. When the PivotTable layout changes, the PivotChart changes with it.

The most important contrast is that a PivotTable **summarises** the source data; it does not replace it and does not change it. If the source data is edited, the PivotTable must be **refreshed** to show the new result. Deleting the source range breaks the PivotTable, and a PivotTable is never the same thing as an Excel table.

27.7 Printing worksheets

A worksheet is printed from **File > Print**, and its layout is controlled from the **Page Layout** tab. The key settings are:

- **Orientation** — **Portrait** for a tall page or **Landscape** for a wide one;
- **Paper size, margins, and headers/footers**;
- **Print Area** — select a range and choose **Set Print Area** to print only that part of the sheet;
- **Print Titles** — repeats chosen header rows or columns at the top or left of every printed page, so a long table keeps its headings; and
- **Scaling** — options such as “Fit Sheet on One Page” shrink the printout to a chosen width or number of pages.

Gridlines are the faint cell borders seen on screen. They are **not printed by default**; the **Print gridlines** option under Sheet Options must be enabled to show them on paper. **Page breaks** can be inserted to control where a new printed page begins.

Example 27.7.1. *A 200-row price list is being printed. Without Print Titles, only the first page shows the column headings and later pages are unreadable. Setting the heading row as a print title repeats it on every page. If the list is a little too wide, landscape orientation or “Fit All Columns on One Page” keeps it tidy.*

27.8 Protecting worksheets and workbooks

Protection restricts changes that can be made to a workbook. **Protect Sheet** (**Review > Protect Sheet**) prevents cells on that sheet from being edited, while allowing selected actions such as selecting locked cells or formatting. **Protect Workbook** protects the **structure** of the workbook — preventing sheets from being added, deleted, renamed, or moved — and can also protect its windows.

Definition (Protection). *A restriction placed on a worksheet or workbook that prevents specified changes, optionally guarded by a password.*

Cell locking is the detail most candidates miss. Every cell is **Locked** by default in the Format Cells dialog, but this lock has **no effect until the sheet is protected**. To let a user edit some cells and not others, the chosen cells must first be **unlocked**, and then the sheet must be protected.

Protection is not the same as encryption. Protection restricts editing inside Excel; it is not the password that encrypts the whole file (that is a separate file-level password).

27.9 What-if analysis

What-if analysis answers the question “what would the result be if an input changed?”. Excel provides three tools for this at the depth a JKSSB candidate needs:

Tool	What it does
Goal Seek	Finds the input value needed to reach a target result.
Scenario Manager	Stores and compares several sets of input values.
Data Table	Shows how an output changes as one or two inputs change.

All three are found under **Data > What-If Analysis**. Goal Seek works backwards from a target: for example, it finds the selling price that makes the profit exactly a desired figure. Scenario Manager keeps named alternatives such as “Best case” and “Worst case”. A Data Table is a grid that shows a range of results for one or two changing inputs.

27.10 Near-neighbour contrast: COUNTIFS vs SUMPRODUCT

To **count rows that satisfy several conditions at the same time**, Excel offers two tools that students often confuse.

COUNTIFS is the direct multi-criterion counting function. Its arguments are written as **paired** ranges and criteria:

COUNTIFS(range1, crit1, range2, crit2, ...)

It applies **AND** logic: a row is counted only if it meets *every* range-and-criteria pair. For example, COUNTIFS(A2:A100, "North", B2:B100, ">500") counts the rows that are in the North region *and* have a value greater than 500.

SUMPRODUCT is a general array function. It can also count multi-criterion rows by multiplying boolean results:

SUMPRODUCT((A2:A100="North")*(B2:B100>500))

Each TRUE/FALSE comparison becomes 1 or 0 when multiplied, and SUMPRODUCT adds the products, giving the same count. Because the multiplication already coerces the boolean values to numbers, SUMPRODUCT does not need the `—` (double unary) that is sometimes seen when boolean arrays are fed to other functions.

For **OR** logic — a row that meets either of two conditions — the two matches are *added*, for example `COUNTIF(range,"North")+COUNTIF(range,"South")`.

Example 27.10.1. *A marksheet has a Stream column and a Marks column. To count students who are in the Science stream and scored more than 80, write*

`COUNTIFS(Stream,"Science",Marks,">80")`.

The same count can be obtained with

`SUMPRODUCT((Stream="Science")*(Marks>80))`.

Confusing the paired arguments of COUNTIFS, or thinking COUNTIFS cannot handle two conditions at once, is the classic error (TRAP-14, PYQ-15).

27.11 Exam retrieval and common confusions

Six distinctions prevent most errors on this topic.

1. **Sort** rearranges rows into an order; **filter** shows only matching rows and **hides** the rest — it does not delete them.
2. A **PivotTable** summarises the source data but is **not** the source table; it must be **refreshed** after the data changes.
3. A PivotTable is **not** an Excel table. The table is the source; the PivotTable is a summary built on it.
4. **COUNTIFS** counts with AND across **paired** range-and-criteria arguments; **SUMPRODUCT** does the same with boolean arrays (TRAP-14).
5. **Locked** cells are protected only **after** the sheet is protected; the default lock has no effect on its own.
6. **Gridlines** are **not printed** unless Print gridlines is enabled.

Example 27.11.1. *An item states, “A PivotTable changes the source data and updates on its own.” Both halves are wrong. A PivotTable never changes the source; and after the source changes it does not update by itself — the user must refresh it. The stable concept is: PivotTable = summary, source = untouched, refresh = manual.*

Exercises

1. Distinguish between sorting and filtering, and state what happens to the rows that a filter does not show.
2. Name two features that an Excel table provides but a plain range does not.

3. Match each chart to its best use: column, line, pie, scatter.
4. List the four areas of a PivotTable and say what goes into each.
5. Explain why a PivotTable must be refreshed after the source data changes.
6. What is the difference between Protect Sheet and Protect Workbook? When does a locked cell actually become protected?
7. Write the COUNTIFS formula that counts rows in the North region with a value greater than 500, and give the equivalent SUMPRODUCT formula.

Solutions

1. Sorting rearranges rows into a chosen order; filtering shows only the rows matching a condition. A filter *hides* the other rows temporarily; it does not delete them, and clearing the filter restores them.
2. An Excel table has a name and auto-expands when new rows are typed; it also provides structured references, banded rows, and built-in filter buttons. A plain range has none of these by default.
3. A column chart compares values across categories; a line chart shows a trend over time; a pie chart shows parts of one whole; a scatter chart shows the relationship between two numeric variables.
4. Rows — field values down the left; Columns — field values across the top; Values — the numbers that are summarised; Filters — fields used to filter the whole PivotTable.
5. A PivotTable stores only a summary view; it does not track the source automatically. After the source data changes, the user must Refresh the PivotTable so the summary is recalculated.
6. Protect Sheet stops cells on a sheet from being edited; Protect Workbook protects the workbook structure (adding, deleting, renaming, or moving sheets). A cell is Locked by default, but the lock takes effect only after the sheet is protected.
7. COUNTIFS(A2:A100,"North", B2:B100,">500"); the equivalent is SUMPRODUCT((A2:A100="North")*(B2:B100>500)).